

Standard Output in C++ Using `cout`

1. What is Standard Output?

In C++, **standard output** is used to display information on the computer screen.

The standard output object in C++ is:

```
cout
```

`cout` is provided by the `<iostream>` library.

We commonly use `cout` to display:

- Text
- Numbers
- Variables
- Calculation results
- Multiple values

2. Basic Syntax of `cout`

The basic syntax is:

```
cout << "Your message here";
```

Complete Program

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello World!";

    return 0;
}
```

Output

```
Hello World!
```

3. Understanding the Code

```
#include <iostream>
```

```
#include <iostream>
```

Includes the **input/output stream library**.

It provides objects such as `cout` and `cin`.

`cout`

```
cout
```

`cout` is the **standard output object** used to display information on the screen.

`<<` Insertion Operator

```
cout << "Hello World!";
```

The `<<` symbol is called the **insertion operator**[1].

It sends the data on its right side to `cout`, which displays it on the screen.

Think of it as:

```
Data → cout → Screen
```

4. Printing Text

We can use `cout` to display text.

Example

```
cout << "Hello World!";
```

Output

```
Hello World!
```

Text must be written inside **double quotation marks** (" ").

5. Printing Numbers

`cout` can also display numbers.

Example

```
cout << 100;
```

Output

```
100
```

We do not use quotation marks when displaying a number.

Compare:

```
cout << 100;
```

Output:

```
100
```

Whereas:

```
cout << "100";
```

also displays:

```
100
```

However, the first is a **number**, while the second is **text**.

6. Printing Variables

`cout` can be used to display the value stored in a variable.

Example

```
int a = 10;  
  
cout << a;
```

Output

```
10
```

We do not put the variable name inside quotation marks.

Example with Text

```
int a = 10;  
  
cout << "Value of a is: " << a;
```

Output

```
Value of a is: 10
```

Understand the Code

```
cout << "Value of a is: " << a;
```

There are two things being displayed:

1. "Value of a is: " → text
2. `a` → value stored in the variable

The insertion operator `<<` connects them.

7. Printing Multiple Values

We can use multiple `<<` operators in one `cout` statement.

Example

```
int age = 20;
float marks = 85.5;

cout << "Age: " << age << endl;
cout << "Marks: " << marks;
```

Output

```
Age: 20
Marks: 85.5
```

8. Printing Multiple Lines

We can use `endl` to move the cursor to the next line.

Example

```
cout << "Line 1" << endl;
cout << "Line 2";
```

Output

```
Line 1
Line 2
```

We can also use the newline escape sequence `\n`:

```
cout << "Line 1\n";
cout << "Line 2";
```

Both `endl` and `\n` can be used to move to the next line.

9. `cout` with Input and Calculation

`cout` is often used together with `cin`.

- `cin` → takes input from the user.
- `cout` → displays output to the user.

Think of the process as:

User Input → cin → Variables → Calculation → cout → Screen

Example 1: Add Two Numbers

Question

Write a C++ program that takes two floating-point numbers from the user, adds them, and displays the sum.

Program

```
#include <iostream>
using namespace std;

int main() {

    float num1, num2, sum;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    sum = num1 + num2;

    cout << "The sum is: " << sum << endl;

    return 0;
}
```

Sample Output

```
Enter first number: 12.5
Enter second number: 7.5
The sum is: 20
```

What This Program Does

1. Declares three `float` variables: `num1`, `num2`, and `sum`.
2. Takes the first number from the user.
3. Takes the second number from the user.
4. Adds the two numbers.
5. Stores the result in `sum`.
6. Uses `cout` to display the result.

Important Line

```
cout << "The sum is: " << sum << endl;
```

Here:

- "The sum is: " → displays text.
- sum → displays the calculated value.
- endl → moves the cursor to the next line.

Example 2: Calculate the Area of a Rectangle

Question

Write a C++ program that takes the height and width of a rectangle from the user, calculates its area using:

```
Area = Height × Width
```

and displays the result.

Program

```
#include <iostream>
using namespace std;

int main() {

    float height, width, area;

    cout << "Enter height of the rectangle: ";
    cin >> height;

    cout << "Enter width of the rectangle: ";
    cin >> width;

    area = height * width;

    cout << "The area of the rectangle is: " << area << endl;

    return 0;
}
```

Sample Output

```
Enter height of the rectangle: 5.5
Enter width of the rectangle: 4
```

```
The area of the rectangle is: 22
```

What This Program Does

1. Declares `height`, `width`, and `area` as `float` variables.
2. Takes the height from the user.
3. Takes the width from the user.
4. Calculates the area:

```
area = height * width;
```

5. Displays the calculated area using `cout`.

10. Using `std::cout`

We can also use `cout` without writing:

```
using namespace std;
```

In that case, we write:

```
std::cout << "Hello World!";
```

Example

```
#include <iostream>

int main() {

    std::cout << "Hello World!";

    return 0;
}
```

Here, `std::cout` means that `cout` belongs to the **standard (std) namespace**.

For beginners, we commonly use:

```
using namespace std;
```

so that we can simply write:

```
cout
```

instead of:

```
std::cout
```

11. Quick Review

Component	Purpose
<code>#include <iostream></code>	Provides input/output features
<code>cout</code>	Displays output on the screen
<code><<</code>	Insertion operator; sends data to <code>cout</code>
<code>endl</code>	Moves the cursor to the next line
<code>\n</code>	Moves the cursor to the next line
<code>cin</code>	Takes input from the user
<code>std::cout</code>	<code>cout</code> from the standard namespace

Key Points to Remember

`cout`

Used to **display output**.

```
cout << "Hello";
```

`<<`

Called the **insertion operator**.

```
cout << value;
```

Text

Write text inside double quotation marks:

```
cout << "Hello World!";
```

Variable

Do not use quotation marks around a variable:

```
int age = 20;  
  
cout << age;
```

Text + Variable

Use multiple `<<` operators:

```
cout << "Age: " << age;
```

New Line

Use either:

```
cout << "Hello" << endl;
```

or:

```
cout << "Hello\n";
```

Practice Questions

Practice 1 – Student Information

Write a C++ program that takes the student's **name, age, and marks** as input and displays them using `cout`.

Practice 2 – Calculate Total

Write a C++ program that takes the prices of three items from the user, calculates the total, and displays the result.

Practice 3 – Calculate Rectangle Perimeter

Write a C++ program that takes the length and width of a rectangle and calculates:

```
Perimeter = 2 × (Length + Width)
```

Display the result using `cout`.

Practice 4 – Calculate Average

Write a C++ program that takes marks of three subjects, calculates the average, and displays the result.

Remember

```
cin  → Input  → User → Program  
cout → Output → Program → Screen
```

cin takes data from the user, while **cout** displays data on t

References and Bibliography

- [1]“operator overloading - cppreference.com,” Cppreference.com, 2026.
<https://en.cppreference.com/cpp/language/operators> (accessed Oct. 06, 2026).